

Figure 5.3 The traditional pie chart (bottom right) represents proportion as an angled slice of a circle. With slight modification, it can be turned into a donut or ring chart (top) or an exploded pie chart (bottom left).

Our `pieChart` function created a new array of three objects. The `startAngle` and `endAngle` for each of the data values draw a pie chart with one piece from 0 degrees to pi, the next from pi to 1.5 pi, and the last from 1.5 pi to 2 pi. But this isn't a drawing, or SVG code like the line and area generators produced.

```
var pieChart = d3.layout.pie();
undefined
var yourPie = pieChart([1,1,2]);
undefined
console.log(yourPie)
▼ [Object, Object, Object]
  ▼ 0: Object
      data: 1
      endAngle: 4.71238898038469
      startAngle: 3.141592653589793
      value: 1
    ▶ __proto__: Object
  ▼ 1: Object
      data: 1
      endAngle: 6.283185307179586
      startAngle: 4.71238898038469
      value: 1
    ▶ __proto__: Object
  ▼ 2: Object
      data: 2
      endAngle: 3.141592653589793
      startAngle: 0
      value: 2
    ▶ __proto__: Object
    length: 3
  ▶ __proto__: Array[0]
```

Original dataset
A layout takes one (and sometimes more) datasets. In this case, the dataset is an array of numbers [1,1,2]. It transforms that dataset for the purpose of drawing it.

Transformed dataset
The layout returns a dataset that has a reference to the original data but also includes new attributes that are meant to be passed to graphical elements or generators. In this case, the pie layout creates an array of objects with the endAngle and startAngle values necessary for the arc generator to create the pie pieces necessary for a pie chart.

Figure 5.4 A pie layout applied to an array of [1,1,2] shows objects created with a start angle, end angle, and value attribute corresponding to the dataset, as well as the original data, which in this case is a number.